

- 4 (a) (i) From Fig. 4.2, amplitude decreases with distance from the source. **B1**
 Since intensity (or power) is proportional to (amplitude)², **B1**
 wave is losing power as it moves away from the source.

MC:	When symbols were used in the explanation, there is a need to define the meaning of the symbols e.g. $I \propto A^2$ will not be accepted unless the meaning of I and A is explained.
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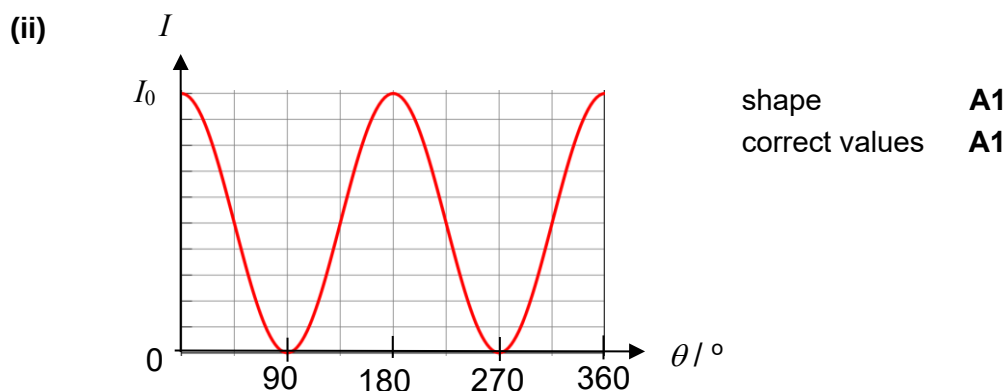
(ii) intensity $\propto$ (amplitude)²

$$\text{ratio} = \frac{(2.0)^2}{(1.1)^2} \quad \text{M1}$$

$$= 3.3 \quad \text{A1}$$

MC:	Students need to evaluate the ratio and not leave in fraction or ratio form.
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(b) (i) $I = I_0 \cos^2 \theta$ **A1**



MC:	While it is a sketch, students should also pay attention to the correct shape of the $\cos^2 \theta$ graph, especially its turning points.
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(iii) 1 After passing through A, intensity is still I_0 , and after passing B,

$$I_{AB} = I_0 \cos^2 45^\circ \quad \text{A1}$$

$$= 0.5 I_0 \quad \text{A1}$$

2 After passing through B, intensity becomes $0.5 I_0$, and after passing through A, intensity $I_{BA} = 0.5 I_0 \cos^2 45^\circ$ **C1**

$$= 0.25 I_0 \quad \text{A1}$$

MC:	Never leave the final answer in fraction form.
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5 (a) (i) This increases the magnetic flux linkage with secondary coil **P1**